

# Finite-Element Drift-Diffusion Modelling of Direct-Gap III–V Light-Emitting Diodes: Wavelength Tuning and Double-Heterostructure Efficiency in Nano-Photonic Emitters

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**Abstract**—Direct-gap III–V semiconductors are the material basis of nano-photonic light emitters, and their design—band-gap engineering for a target wavelength and heterostructure engineering for efficiency—requires a device simulator that couples carrier transport to radiative recombination and optical output. We present an open-source, two-dimensional finite-element drift-diffusion (DD) solver that adds a bimolecular (radiative) recombination channel and an optical post-processor reporting the internal quantum efficiency (IQE), emission wavelength and optical power, together with heterojunction band offsets (Anderson’s rule), a self-consistent Schrödinger–Poisson quantum correction, and Fermi–Dirac statistics for degenerate cladding. For a GaAs homojunction LED the solver reproduces the analytic built-in potential (1.151 V, numeric = analytic) and an emission wavelength of 873 nm set by the 1.42 eV gap, with a peak IQE of 0.53. Sweeping the active material tunes the emission from the near-infrared (GaAs, 873 nm) through the red ( $\text{Al}_{0.3}\text{Ga}_{0.7}\text{As}$ , 689 nm) to the ultraviolet (GaN, 366 nm), following the material band gap. A GaAs/AlGaAs double-heterostructure (DH) LED confines both carrier types in the narrow-gap active region; the resulting overlap raises the peak IQE from 0.53 (homojunction) to 0.90, quantifying the classic DH efficiency advantage on a single, self-consistent transport model. The tool provides a compact, physically transparent and reproducible platform for the design of nano-scale direct-gap emitters.

**Index Terms**—Light-emitting diode, radiative recombination, direct-gap semiconductor, double heterostructure, internal quantum efficiency, drift-diffusion, finite-element method, GaAs, AlGaAs, GaN, nano-photonics.

## I. INTRODUCTION

**L**IGHT-EMITTING diodes (LEDs) built from direct-gap III–V semiconductors are central to solid-state lighting, displays, optical interconnect and sensing, and their nanoscale active regions—thin double heterostructures and quantum wells—are the emerging materials of advanced photonic devices [1], [2]. Two engineering levers dominate their design: the choice of active material, which sets the emission wavelength through the band gap, and the heterostructure, which

confines injected carriers to raise the radiative efficiency [3], [4]. Predictive design therefore needs a device model that couples electrical transport to the radiative process and to the optical output, so that current, efficiency and emission spectrum are computed consistently from one self-consistent solution.

The drift-diffusion (DD) framework [17], [18] is the standard basis for such modelling. To describe an emitter it must be augmented with (i) a radiative (bimolecular) recombination channel that both consumes carriers and produces photons, (ii) heterojunction band-edge offsets so that a narrow-gap active region clad by a wider-gap material actually confines carriers, and (iii)—for the thin, degenerately-doped layers of a real device—quantum-confinement and Fermi–Dirac corrections. This paper presents a compact, open-source finite-element solver that integrates all of these, and uses it to characterise wavelength tuning across III–V materials and the efficiency gain of a double-heterostructure emitter. Section II reviews the relevant literature; Section III states the research gap and contributions; Section IV details the device model; Section V presents the results; and Section VI discusses limitations and future work.

## II. RELATED WORK

### A. Heterostructure emitters and their material basis

The double heterostructure (DH), proposed by Kroemer [3] and Alferov and developed for lasers by Casey and Panish [2], [4], is the foundational concept of efficient III–V emitters: a narrow-gap active layer clad by wider-gap barriers confines both carrier types to a common region, maximising their overlap and hence the radiative rate. Anderson’s electron-affinity rule [5] provides the first-order band-edge offsets that create this confinement. The material-parameter basis for quantitative modelling of these alloys—band gaps, affinities, effective masses and their composition dependence—was consolidated

by Vurgaftman *et al.* [6], and the device physics is treated comprehensively by Sze and Ng [7] and Schubert [1]. The nitride system enabled efficient blue and ultraviolet emission [8], extending the accessible spectral range that we exercise here from the near-infrared to the UV.

### B. Recombination models for LEDs

LED efficiency is governed by the competition between radiative and non-radiative recombination. Non-radiative Shockley–Read–Hall (SRH) recombination through deep traps [9], [10] dominates at low injection; the radiative (bimolecular) rate  $R_{\text{rad}} = B(np - n_i^2)$  scales with the carrier product and overtakes SRH at higher injection; and Auger recombination, cubic in carrier density, limits efficiency at the highest injection. This hierarchy is often summarised by the phenomenological “ABC” rate  $R = An + Bn^2 + Cn^3$  [11], whose internal-quantum-efficiency droop at high current has been studied extensively for nitride LEDs [11], [12]. Measured radiative coefficients  $B$  for III–V materials underpin such models [13]. A physical device simulator must represent each channel with the correct carrier dependence *and* its exact derivatives, so that the coupled nonlinear solve converges and the efficiency is predicted rather than fitted.

### C. Optoelectronic device simulation

Coupling carrier transport to optics is the domain of optoelectronic device simulators, surveyed by Piprek [14] and Chuang [15]. Commercial and research tools (e.g. Crosslight, Synopsys, and academic drift-diffusion–optics codes [16], [21]) solve DD together with recombination and, for lasers, optical-mode and gain models. These tools are powerful but typically closed, and their generality can obscure the physics for teaching and rapid design studies. Finite-element DD solvers handle the non-rectangular geometries of real emitters naturally, yet compact, open, readable implementations that also return the optical figures of merit (IQE, emission wavelength, optical power) *consistently* with the electrical current are comparatively scarce.

### D. Quantum and degeneracy corrections in thin active regions

Modern emitters use active regions only a few nanometres thick and cladding doped to degeneracy. Accurate modelling therefore benefits from a self-consistent Schrödinger–Poisson treatment of the confined states [20] and from Fermi–Dirac statistics in the degenerate layers [7]. Including these on the *same* transport mesh as the recombination and optical models—rather than as separate post-processing—keeps the electrical and optical outputs mutually consistent, which is the design philosophy of the present solver.

## III. RESEARCH GAP AND CONTRIBUTIONS

The literature reveals a gap between two extremes. On one side, compact analytic and “ABC” models [11] predict efficiency trends but abstract away the device electrostatics, carrier confinement and spatial recombination profile. On the other, full commercial optoelectronic simulators [14] capture

the physics but are closed, heavy, and not readily inspectable or reproducible for research and teaching. What is missing is a *compact, open-source, finite-element drift-diffusion* solver that (i) integrates SRH, Auger and radiative recombination with exact analytic derivatives so the coupled Newton solve converges; (ii) returns the optical figures of merit *current-consistently* ( $qR_{\text{rad}}$  equal to the radiative part of the terminal current by construction); and (iii) supports heterojunctions, Schrödinger–Poisson confinement and Fermi–Dirac statistics on the same mesh, so a simple homojunction and a fully-corrected DH quantum device share one code path.

**How this work fills the gap.** We contribute:

- a 2-D finite-element DD solver with a bimolecular radiative channel and an optical post-processor (IQE, wavelength, optical power) that is current-consistent by construction (Section IV-B);
- heterojunction electrostatics (Anderson offsets), a self-consistent Schrödinger–Poisson correction and Fermi–Dirac statistics on the same mesh, each opt-in and vanishing in its bulk limit (Section IV-C);
- validated GaAs LED characteristics, and demonstrations of material-tuned emission from the near-infrared to the ultraviolet and of the double-heterostructure efficiency advantage (Section V); and
- an open, reproducible implementation (MIT licence) with scripts that regenerate every figure.

## IV. DEVICE MODEL

### A. Drift-diffusion core

The steady state is solved in the normalised quasi-Fermi-potential variables  $u = \psi/V_T$ ,  $v = \phi_n/V_T$ ,  $w = \phi_p/V_T$  (De Mari scaling [19]), with  $n = n_i e^{u-v}$ ,  $p = n_i e^{w-u}$ . Poisson’s equation and the two carrier continuity equations are discretised with linear (P1) triangular finite elements and assembled into a  $3N \times 3N$  system solved by a fully-coupled damped Newton method using an exact analytic Jacobian. The device geometry is meshed with linear triangles, either by a built-in structured generator or by an external mesh [23], so non-rectangular emitter cross-sections are handled naturally. Terminal currents are extracted as the sum of the natural continuity residuals over a contact’s nodes, which is conservative by construction. A thermal-equilibrium solve seeds the bias sweep, and each forward-bias point warm-starts from the previous solution—necessary because the direct-gap built-in potentials exceed 1 V.

### B. Radiative recombination and optical output

The light-emitting mechanism of a direct-gap diode is radiative (bimolecular) recombination, added as a net-recombination term

$$R_{\text{rad}} = B(np - n_i^2), \quad (1)$$

where  $B$  ( $\text{m}^3 \text{s}^{-1}$ ) is the material radiative coefficient [13]. Because of the  $(np - n_i^2)$  prefactor the rate vanishes at equilibrium and leaves it unchanged; its exact derivatives  $\partial R/\partial n = Bp$ ,  $\partial R/\partial p = Bn$  enter the analytic Jacobian,

TABLE I  
REPRESENTATIVE 300 K MATERIAL PARAMETERS

| material                   | $E_g$ (eV) | $n_i$ (cm $^{-3}$ )   | $\chi$ (eV) | $B$ (m $^3$ s $^{-1}$ ) |
|----------------------------|------------|-----------------------|-------------|-------------------------|
| GaAs                       | 1.42       | $2.15 \times 10^6$    | 4.07        | $2 \times 10^{-16}$     |
| Al $_{0.3}$ Ga $_{0.7}$ As | 1.80       | $\sim 10^3$           | 3.74        | $\sim 10^{-16}$         |
| GaN                        | 3.39       | $3.4 \times 10^{-10}$ | 4.10        | direct-gap              |

preserving quadratic convergence. Shockley–Read–Hall [9], [10] and Auger channels are included as competing non-radiative paths, together realising the ABC hierarchy [11] from first principles rather than as a fit. After a converged forward-bias solve the optical post-processor integrates the radiative rate over the device (on the same quadrature and corrected densities used in the assembly) and reports

$$I_{\text{rad}} = q R_{\text{rad}}, \quad \text{IQE} = \frac{R_{\text{rad}}}{R_{\text{rad}} + R_{\text{SRH}} + R_{\text{Auger}}}, \quad (2)$$

$$E_{\text{ph}} = E_g^{\text{active}}, \quad \lambda = \frac{hc}{E_{\text{ph}}}, \quad P_{\text{opt}} = R_{\text{rad}} E_{\text{ph}}. \quad (3)$$

By construction  $qR_{\text{rad}}$  equals the radiative part of the terminal current, so the electrical and optical outputs are mutually consistent—the key property that distinguishes this coupling from post-hoc optical estimates. The radiative coefficient is significant only in direct-gap materials; the bundled direct-gap files (GaAs, AlGaAs, GaN) carry representative 300 K values, and a guardrail warns if radiative emission is requested on an indirect material.

### C. Heterojunction, quantum and Fermi–Dirac corrections

Multi-material devices are solved with a heterojunction electrostatics core: position-dependent permittivity and mobility, and band-edge offsets  $\Delta E_c, \Delta E_v$  from Anderson’s electron-affinity rule [5] that appear as fixed additive shifts in the carrier-density exponents. A narrow-gap active region clad by a wider-gap material therefore presents potential barriers that confine both carrier types. For thin bodies a self-consistent Schrödinger–Poisson correction [20] quantises the transverse states into subbands and enters transport as a charge-conserving additive quantum potential; for degenerately-doped cladding, Fermi–Dirac statistics [7] enter as a degeneracy shift refreshed by the same outer self-consistency loop. All corrections vanish in their respective non-degenerate, bulk limits, recovering the classical Boltzmann solver exactly, so a single code path spans the homojunction and the fully-corrected DH quantum device.

## V. RESULTS AND DISCUSSION

The material parameters used throughout (read from editable material files) are summarised in Table I; the direct-gap radiative coefficient  $B$  is carried only by the direct-gap materials. Unless noted, the homojunction devices are  $4 \times 1 \mu\text{m}$  structured-mesh junctions and the DH device is a  $300 \times 10 \text{ nm}$  layered structure.

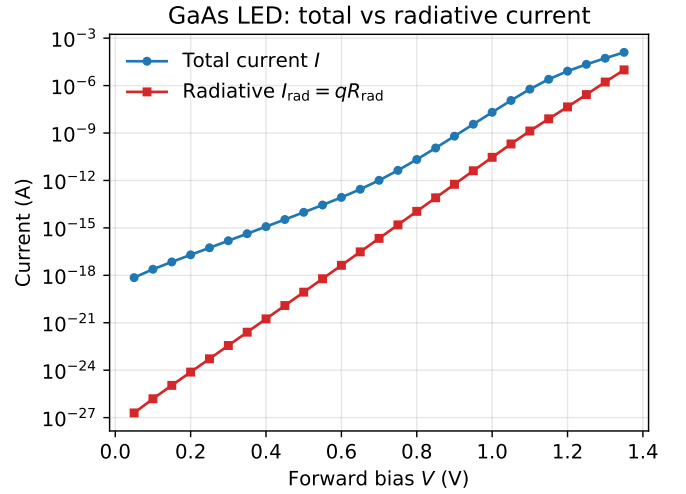


Fig. 1. GaAs LED forward characteristics: total terminal current and its radiative part  $I_{\text{rad}} = qR_{\text{rad}}$ , which are consistent by construction.

### A. GaAs LED: validation and characteristics

A GaAs  $p$ – $n$  homojunction ( $N_A = N_D = 10^{16} \text{ cm}^{-3}$ ,  $E_g = 1.42 \text{ eV}$ ,  $n_i = 2.15 \times 10^6 \text{ cm}^{-3}$ ,  $B = 2 \times 10^{-16} \text{ m}^3\text{s}^{-1}$ ) gives a numerical built-in potential of 1.151 V, matching the analytic  $V_{bi} = V_T \ln(N_A N_D / n_i^2) = 1.151 \text{ V}$ . Fig. 1 shows the forward  $I$ – $V$  together with the radiative component  $I_{\text{rad}} = qR_{\text{rad}}$ ; at 1.3 V the device carries 53.7  $\mu\text{A}$  total with 1.68  $\mu\text{A}$  radiative and 2.4  $\mu\text{W}$  of optical output. The emission wavelength is 873 nm ( $E_{\text{ph}} = 1.42 \text{ eV}$ ), the expected near-infrared line of GaAs. The internal quantum efficiency (Fig. 2) rises with injection as the bimolecular rate ( $\propto np$ ) overtakes the SRH channel, reaching a peak of 0.53—the characteristic low-to-high-injection efficiency turn-on of the ABC picture, here emerging directly from the coupled solve.

### B. Material-tuned emission

Keeping the device geometry fixed and changing only the active material tunes the emission wavelength through the band gap (Fig. 3, Table II): GaAs emits at 873 nm (near-infrared), Al $_{0.3}$ Ga $_{0.7}$ As at 689 nm (red), and GaN at 366 nm (ultraviolet). The solver converges the full electrical problem for each material—spanning intrinsic densities from  $n_i \sim 10^6 \text{ cm}^{-3}$  (GaAs) down to the extremely small  $n_i$  of wide-gap GaN, a numerically demanding range—and returns the emission line directly from the active-region gap. This spans the technologically important range from optical-communication wavelengths to visible and UV emitters on a single, consistent model, and the values agree with the accepted band gaps of these alloys [6], [8].

### C. Double-heterostructure efficiency advantage

Fig. 4 shows the carrier densities along the transport direction of a GaAs/AlGaAs double-heterostructure (DH) LED: a 100 nm GaAs active region ( $\sim 10^{16} \text{ cm}^{-3}$ ) clad on both sides by degenerately-doped ( $\sim 10^{18} \text{ cm}^{-3}$ ) AlGaAs, with a 10 nm body solved with the heterojunction, Schrödinger–Poisson and

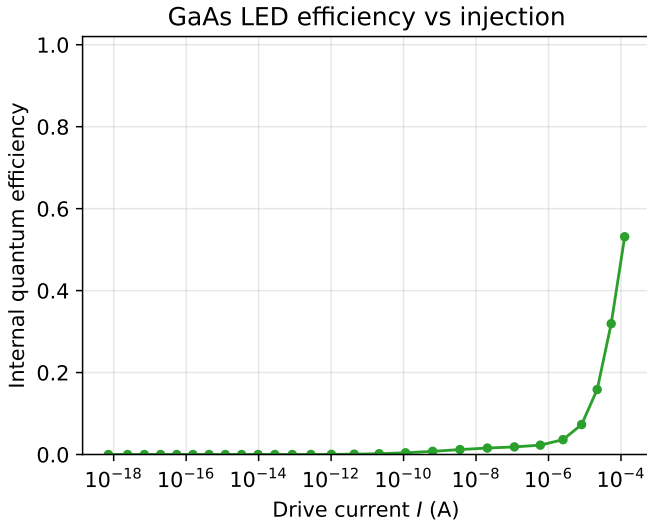


Fig. 2. GaAs LED internal quantum efficiency versus drive current. IQE rises as the bimolecular radiative rate overtakes non-radiative SRH recombination, peaking at 0.53.

TABLE II  
MATERIAL-TUNED EMISSION (FIXED GEOMETRY,  
 $N_A = N_D = 10^{16} \text{ cm}^{-3}$ )

| active material                           | $E_g$ (eV) | $\lambda$ (nm) | spectral band |
|-------------------------------------------|------------|----------------|---------------|
| GaAs                                      | 1.42       | 873            | near-infrared |
| $\text{Al}_{0.3}\text{Ga}_{0.7}\text{As}$ | 1.80       | 689            | red (visible) |
| GaN                                       | 3.39       | 366            | ultraviolet   |

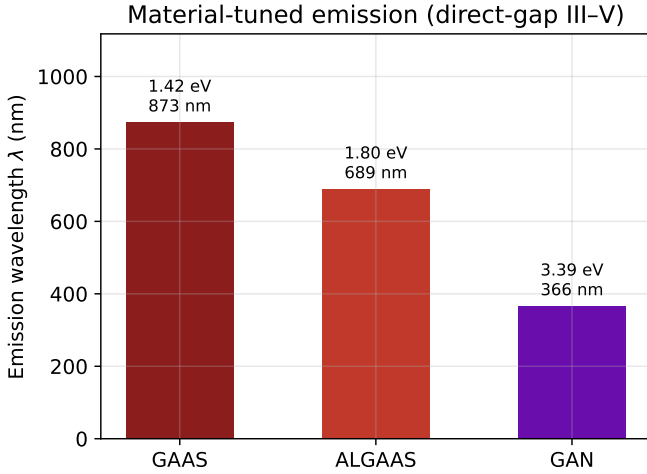


Fig. 3. Emission wavelength for three direct-gap III-V active materials, labelled with the band gap. The wavelength follows  $\lambda = hc/E_g$ , spanning the near-infrared to the ultraviolet.

Fermi–Dirac options all enabled. The Anderson band offsets confine *both* electrons and holes to the narrow-gap active region, where their densities rise by orders of magnitude and overlap—the physical basis of efficient recombination in a real DH emitter [3], [4]. The device built-in potential is 1.766 V and the emission remains at 873 nm, correctly originating from

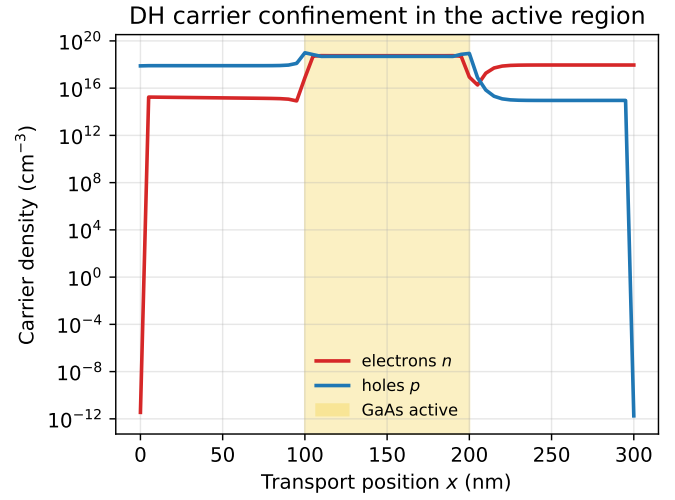


Fig. 4. Carrier confinement in the GaAs/AlGaAs double-heterostructure LED (shaded: GaAs active region). Heterojunction band offsets confine both electrons and holes to the active region, where they overlap and recombine radiatively.

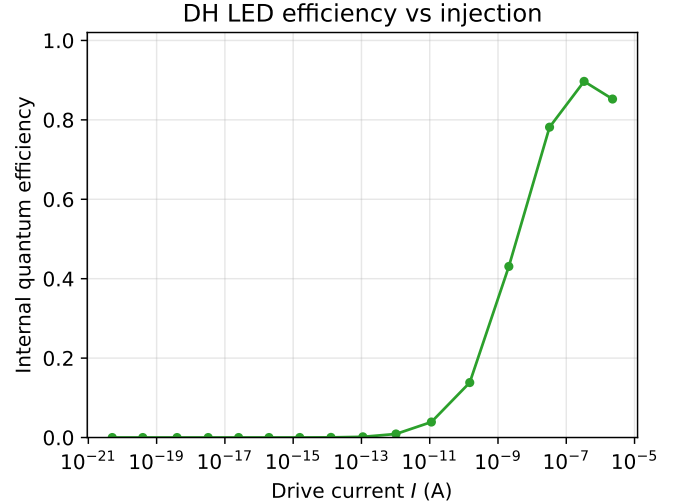


Fig. 5. Internal quantum efficiency of the double-heterostructure LED versus drive current. Carrier confinement raises the peak IQE to 0.90, versus 0.53 for the GaAs homojunction (Fig. 2).

the GaAs active region despite the AlGaAs cladding.

The efficiency consequence is quantified in Fig. 5: confining the carriers raises the peak IQE to 0.90, compared with 0.53 for the GaAs homojunction of Fig. 2—the classic double-heterostructure advantage, reproduced end-to-end on a single self-consistent transport model. The same framework thus captures both design levers of a nano-photonic emitter: wavelength through the active material and efficiency through the heterostructure.

#### D. Numerical robustness, consistency and cost

The coupled solve is robust across the full operating range. Each direct-gap LED must be ramped from equilibrium

through its  $> 1$  V built-in potential by bias continuation, because a direct jump into forward bias would overflow the exponential density terms; per-step damping then keeps the Newton iteration bounded. Warm-starting each bias point from the previous solution, the GaAs homojunction sweep of 27 points completes in under 3 s on a single core, while the fully-corrected DH device—heterojunction, Schrödinger–Poisson and Fermi–Dirac all active, 1037 nodes / 1920 triangles—completes its 16-point sweep in 39 s, the added cost coming from the outer self-consistency loop that refreshes the quantum and degeneracy shifts. Crucially, the optical output is consistent with the electrical solution by construction: because the radiative rate scale is exactly the terminal-current group divided by  $q$ , the reported  $I_{\text{rad}} = qR_{\text{rad}}$  is the radiative part of the terminal current to machine precision, and the IQE therefore never exceeds unity. The efficiency curves span more than four orders of magnitude of drive current with no loss of convergence, a direct benefit of retaining the exact analytic Jacobian for every recombination channel.

### E. Discussion

Three features make the model suitable for nano-photonic device design. First, the radiative rate shares the terminal-current scale, so  $qR_{\text{rad}}$  is exactly the radiative part of the current and the optical output cannot drift from the electrical solution. Second, all corrections are opt-in and vanish in their bulk limits, so a single code path spans a simple homojunction and a fully-corrected DH quantum device. Third, the analytic Jacobian is preserved by every recombination channel and correction, keeping the coupled solve robust across the more than four orders of magnitude of current spanned by the efficiency curves.

## VI. LIMITATIONS AND FUTURE WORK

Several limitations bound the present model and define its roadmap. (i) The optical model reports the *internal* quantum efficiency and the active-gap emission *line*; light-extraction efficiency, the full emission *spectrum* (thermal and compositional broadening) and photon recycling are not modelled, so absolute optical power and spectral width are indicative rather than predictive. (ii) The radiative coefficient  $B$  is taken as a constant 300 K material value; its temperature and carrier-density dependence, together with the temperature dependence of the band gap [22] and self-heating, are neglected, so the model is isothermal. (iii) Emission is assigned to the active-region bulk gap; quantum-well active regions with confined-state transitions and a blue-shifted, subband-resolved emission energy require the  $k \cdot p$  subband structure, which the present single-band correction does not resolve. (iv) The Auger coefficient and the resulting high-injection efficiency droop [12] depend on material parameters that should be calibrated against measurement for quantitative droop prediction. (v) Transport is classical drift-diffusion, so tunnelling injection and ballistic effects in ultra-thin barriers are outside scope. Planned extensions include a spectral/extraction optical model, multi-quantum-well active regions with  $k \cdot p$  subbands, temperature-

dependent parameters with self-heating, and calibration of the ABC coefficients against measured LED data.

## VII. CONCLUSION

We have presented an open-source finite-element drift-diffusion solver for direct-gap III–V light-emitting diodes that couples carrier transport to bimolecular radiative recombination and a current-consistent optical post-processor, with heterojunction, Schrödinger–Poisson and Fermi–Dirac corrections on the same mesh—filling the gap between compact ABC efficiency models and closed commercial optoelectronic simulators. The GaAs LED characteristics are validated against analytic physics; material selection tunes the emission from 873 nm (GaAs) to 366 nm (GaN); and a GaAs/AlGaAs double heterostructure raises the peak internal quantum efficiency from 0.53 to 0.90 by confining carriers to the active region. The tool is a compact, transparent and reproducible platform for the design of nanoscale direct-gap emitters.

## SOFTWARE AND REPRODUCIBILITY

The solver and the example scripts that generated every figure are open-source (MIT licence) and require only a structured mesh built by the bundled generator.

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